Sachin Sravan Kumar Komati

AI/ML Engineer

FL | 954-228-3704| sachin.k@itjobinbox.com

GitHub |Portfolio | Google Scholar | Diagnostic Tools |FIU Profile |LinkedIn

SUMMARY

AI/ML Engineer with 5.6+ years of experience delivering production-grade machine learning and GenAI systems in regulated finance and healthcare. Specialized in RAG-based LLM platforms, high-throughput model serving, and cost-efficient RLHF pipelines using Azure OpenAI, vLLM, Kubernetes, and MLflow, driving up to 24× throughput gains and 75% cost reduction. Strong background in deep learning, NLP, and multimodal modeling, with hands-on ownership of end-to-end ML pipelines from data engineering through deployment and governance-ready production.

SKILLS

Methodologies: SDLC, Agile, MLOps, Model Lifecycle Management, Cross-Functional Collaboration
Core Competencies: Machine Learning, Deep Learning, Model Deployment, MLOps, GPU Optimization, Computer Vision
Programming: Python, C++, SQL
Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, LightGBM
Generative AI & NLP: Large Language Models, Hugging Face Transformers, BERT, Prompt Engineering, LangChain
GPU Acceleration: CUDA, TensorRT, Quantization, Inference Optimization
Distributed Computing & Data Engineering: Apache Spark, Airflow, Feature Engineering, ETL Pipelines
Cloud & Deployment: AWS SageMaker, Google Cloud AI Platform, Docker, Kubernetes, MLflow, CI/CD Pipelines,
Databases & Storage: MySQL, PostgreSQL, MongoDB
Visualization & Reporting: Tableau, Power BI, Matplotlib, Seaborn
Tools: Git, Jupyter Notebook, FastAPI, Flask

EXPERIENCE

BlackRock, FL | Mar 2025 – Current | AI/ML Engineer

- Implemented domain adaptation of foundation models for regulated financial use cases, integrating **SEC**, **FINRA**, and **ESMA** content via **RAG**, **LangChain**, and instruction fine-tuning to meet production-quality standards.
- Engineered high-throughput LLM serving using **Azure OpenAI** for managed model inference and **vLLM** for self-hosted open-source models with streaming, **FlashAttention**, and quantization, achieving **10×** inference scaling and **24×** throughput gains while cutting memory footprint by **60%**.
- Developed and deployed **RLHF** pipelines using **DPO** and **PPO** combined with **LoRA** and **QLoRA**, reducing GPU hours and training cost by **75%** while preserving model alignment.
- Enhanced the RAG pipeline by integrating **FAISS** and **Pinecone** for vector retrieval, improving retrieval relevance by **22%** and reducing response latency by **30%** for knowledge-based queries.
- Built data aggregation and preprocessing pipelines handling **8M+** structured and unstructured records using **Python**, **Spark**, **Kafka**, and **PostgreSQL**, improving training data quality and enabling reliable domain-specific model fine-tuning.
- Automated end-to-end ML CI/CD with **Kubernetes**, **MLflow**, and **NVIDIA Triton**, cutting model deployment lead time by **40%** and standardizing reproducible, production-ready releases.

GE HealthCare, Remote | Apr 2024 – Mar 2025 | Machine Learning Research Engineer

- Designed and deployed 11 longitudinal survival-prediction models using **LSTM architectures** on a **4M-record clinical dataset**, achieving sub-0.01 MSE through temporal feature engineering, sequence normalization, and bias–variance optimization across patient cohorts.
- Built and validated **DiabCompSepsAI**, a **Random Forest–based postoperative** complication prediction system leveraging multivariate clinical signals, reaching 94%+ accuracy through feature selection, class-imbalance handling, and hyperparameter tuning.
- Developed multimodal **risk-stratification pipelines** combining imaging data, demographics, and structured **clinical metadata** to classify polyp recurrence into low, moderate, and **high-risk cohorts**, enabling personalized post-procedure monitoring strategies.
- Achieved state-of-the-art performance (92.2% accuracy) on capsule **endoscopy lesion detection** by implementing **Vision Transformer pipelines** with domain-specific augmentations and self-supervised pretraining to improve generalization on limited labeled data.
- Engineered an automated **medical-imaging pipeline** processing **120K+ images** and **81 videos**, spanning ingestion, segmentation, augmentation, and model training, significantly improving cross-dataset robustness of **GI anomaly detection** models.
- Delivered high-precision **pathology** and **morphology** classifiers (99.05% and 94.94% accuracy) using **multimodal attention mechanisms** and autoencoder-based anomaly detection, including a **GI bleeding detection system** achieving a 90% **F1-score** under real-world clinical edge cases.

Virtual Infotech Solution, India | Jan 2019 – July 2022 | Machine Learning

- **Designed and productionized supervised and deep learning models** using PyTorch and TensorFlow, applying Bayesian hyperparameter optimization and k-fold cross-validation to cut experimentation cycles by **30%** while maintaining stable out-of-sample performance.
- **Built LSTM-based forecasting models and synthetic data augmentation pipelines** to mitigate data sparsity, improving prediction stability and reducing error volatility by **40%** in downstream analytical reporting.
- **Re-architected NLP pipelines** using spaCy and Hugging Face Transformers, fine-tuning embeddings for entity extraction and text classification, driving a **35% lift in model accuracy** across production workloads.
- **Engineered distributed data processing and feature pipelines** with Spark, Kafka, and Hadoop to support real-time and batch ML workflows, reducing end-to-end data latency by **50%** and enabling near-real-time inference use cases.

- **Developed low-latency ML inference services** using FastAPI and gRPC, implementing async request handling and Redis caching to cut API response times by **40% under peak traffic**.
- **Operationalized ML deployments at scale** by containerizing models with Docker and Kubernetes, automating CI/CD via GitHub Actions and MLflow, and orchestrating retraining with Airflow - improving release reliability and model stability by **40%**.

EDUCATION

Master of Science in Computer Science **May 2024**
Florida International University, Miami, Florida
Bachelor of Technology in Computer Science & Engineering **May 2019**
Jawaharlal Nehru Technological University Kakinada

CERTIFICATIONS

AI in Healthcare
Biostatistics in Public Health
Epidemiology in Public Health Practice
Responsible Conduct of Research (RCR)
Responsible Conduct of Research (RCR)
Biomedical Human Research Course
Good Clinical Practice (GCP)

PUBLICATIONS & PRESENTATIONS

1) Al-delirium guard: Predictive modeling of postoperative delirium in elderly surgical patients, PLOS One **[View Publication]**
2) A machine learning framework for predicting polyp evolution and malignancy risk: Enhancing personalized surveillance in Gastroenterology, Gastrointestinal Endoscopy. **[View Publication]**
3) Al-enhanced prognostic model for predicting polyp recurrence and guiding post-polypectomy surveillance intervals using the ERCPMP-v5 dataset, Gastrointestinal Endoscopy. **[View Publication]**
4) Automated detection of gastrointestinal bleeding in wireless capsule endoscopy images using autoencoders, Gastrointestinal Endoscopy. **[View Publication]**
5) Vision Transformer-based framework for risk stratification and prognostic assessment in gastrointestinal lesion management, Gastrointestinal Endoscopy**. [View Publication]**
6) DiabCompSepsAI: Advanced predictive modeling of postoperative sepsis and wound infection in diabetes, Critical Care Medicine. **[View Publication]**
7) Al-Delirium Guard: Predictive modeling of postoperative delirium in elderly surgical patients, Critical Care Medicine. **[View Publication]**
8) Predictive modeling of GI disease: GastroEndo-Seq for progression and outcome forecasting, Gastroenterology. **[View Publication]**
9) Al-driven real-time detection, classification, and malignancy risk prediction of colorectal polyps using the ERCPMP-v5 dataset, Gastroenterology. **[View Publication]**
10) Utilizing vision transformer architecture to create a model for automated detection of small bowel lesions in capsule endoscopy, Gastroenterology. **[View Publication]**
11) Predicting mortality in upper gastrointestinal tract cancer using long short-term memory neural networks, Journal of Clinical Oncology. **[View Publication]**
12) Refining Pancreatic Cancer Projections: Mortality Trends Analysis and LSTM Model Predictions, American Journal of Gastroenterology. **[View Publication]**
13) Advancing Lower GI Cancer Projections: Mortality Trends Analysis and LSTM Model Predictions, American Journal of Gastroenterology. **[View Publication]**
14) Hepatobiliary Carcinoma: Analyzing Mortality Trends and Developing an LSTM-Based Predictive Model, American Journal of Gastroenterology. **[View Publication]**
15) Detection & Recognition of Leaf Disease Using Image Processing, International Journal of Research. **[View Publication]**

ACHIEVEMENTS

- Appointed as the *MS Representative* for the Student Advisory Board at Florida International University, standing as one among the **2500** graduate students, advocating the interests and concerns of the entire graduate community.
- **29 out of 30** submitted abstracts accepted for **ACG 2025** (American College of Gastroenterology), demonstrating an exceptional acceptance rate and research impact.
- **13** abstracts accepted for **UEG Week 2025** (United European Gastroenterology), one of the largest & most prestigious international conferences in gastroenterology.
- Received meritorious certificate for outstanding contribution towards the success of *StartupStack Entrepreneurial Summit* hosted by StartupStack Organization.
- Ranked in the top **4.5%** out of 200,000 students in EAMCET exam conducted by the State Govt of Andhra Pradesh.
- Recipient of a **100%** scholarship for the entire *Undergraduate* course.

AWARDS

- Recipient of **Star Research Award** at the Society of Critical Care Medicine's (SCCM) 2025 Critical Care Congress for excellence in critical care research.
- Honored with **3 Presidential Awards** for **ACG 2025**, a distinction given to only **5%** of nearly **6,000** abstracts.
- Recognized for **AASLD 2025 – The Liver Meeting** with **3** abstract acceptances, of which **2** received **Poster of Distinction** honors.